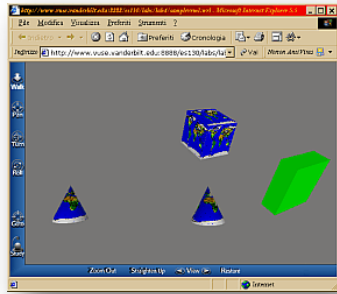


Some Additional Web-Related Topics



Inclusion of MM elements in web pages

In the past: **deprecated** tag `<embed>`

Audio

Example: background music

```
<embed src="s.mid" autostart="true" hidden="true">
```

Video

Example: embedded video

```
<embed src="budapest.avi">
```

The browser needed the proper plugin

A “universal” tag (especially before HTML5)

The <object> tag...

The <object> tag should be used to include elements other than text and images

- With HTML4, the <object> tag was mainly used for audio and video files, Java applets, Flash animations, and ActiveX controls
- It can even be used for web pages or images
Now, much better: for images, <video> and <audio> for multimedia content, and <iframe> for HTML documents
- The value of the `classid` attribute indicates/d the “player” or the plugin to be used to play the multimedia file

A “universal” tag (especially before HTML5)

... the <object> tag

Examples

Video file (old, not used anymore):

```
<object classid="clsid:22D6F312-B0F6-11D0-94AB-0080C74C7E95">  
  <param name="FileName" value="Budapest.avi">  
</object>
```

PDF file (but an <iframe> is more common):

```
<object data="report2025.pdf" type="application/pdf" width="100%"  
  height="100%">  
  <p>  
    Your browser doesn't support embedding PDF files. You can  
    <a href="report2025.pdf">download the file</a> instead.  
  </p>  
</object>
```

The <iframe> tag

An **inline frame** is used to embed another document within the current HTML page

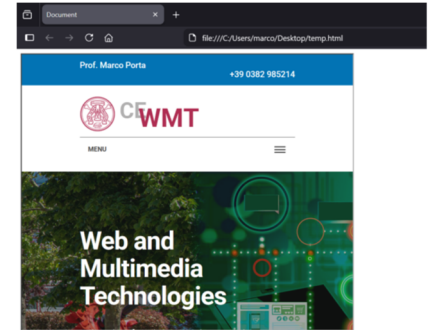
Example:

```
<iframe src="https://vision.unipv.it/wmt/" width="600"
        height="500">
</iframe>
```

However, the web server may block the embedding of a web page...

Example of PDF embedding:

```
<iframe src="report2025.pdf" width="100%" height="100%">
<p>
  It appears your browser does not support iframes, or you can't view PDFs.
  You can <a href="report2025.pdf">download the PDF</a>.
</p>
</iframe>
```



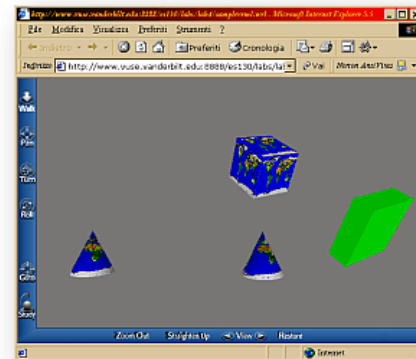
VRML

VRML = Virtual Reality Markup Language

- Old markup language to define “virtual worlds”
- Not XML-based
- Required specific plugins (e.g., CosmoPlayer)

Example of VRML code:

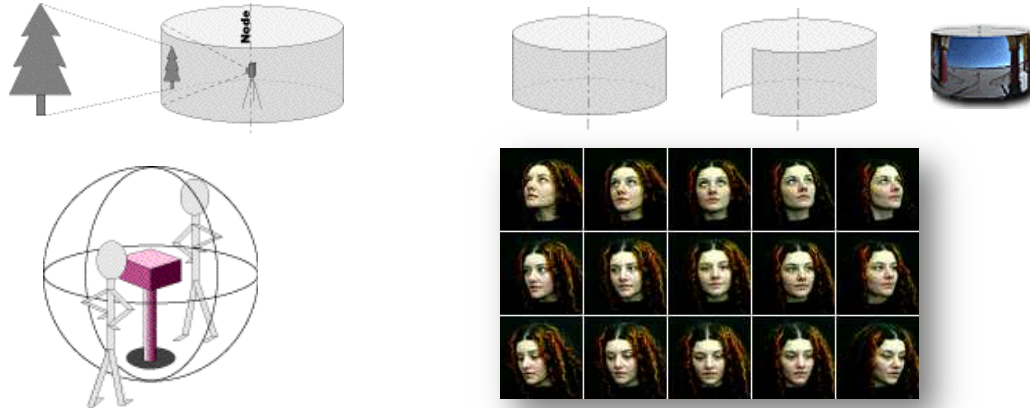
```
.....  
Transform {  
  translation -7 0 0  
  children  
  DEF THISCONE Shape{  
    appearance DEF aper Appearance {  
      material Material {  
        diffuseColor 0 1 0  
      }  
      texture ImageTexture {url "earth.gif"}  
    } geometry Cone { }  
  }  
}  
.....
```



QuickTime VR (QTVR)

QTVR = Quick Time Virtual Reality

Technology created by Apple to build “3D panoramas” from photos (multiple views of the same object)



X3D vs. WebGL

X3D (www.web3d.org)

ISO standard (XML-based) file format for 3D web graphics; is considered the successor of VRML



WebGL (www.khronos.org/webgl/)

The likely “winner”: JavaScript API for rendering interactive 2D and 3D graphics within any compatible web browser without the need for a plugin (it uses the HTML5 <canvas> tag)



Deep and Dark Web...

Deep Web

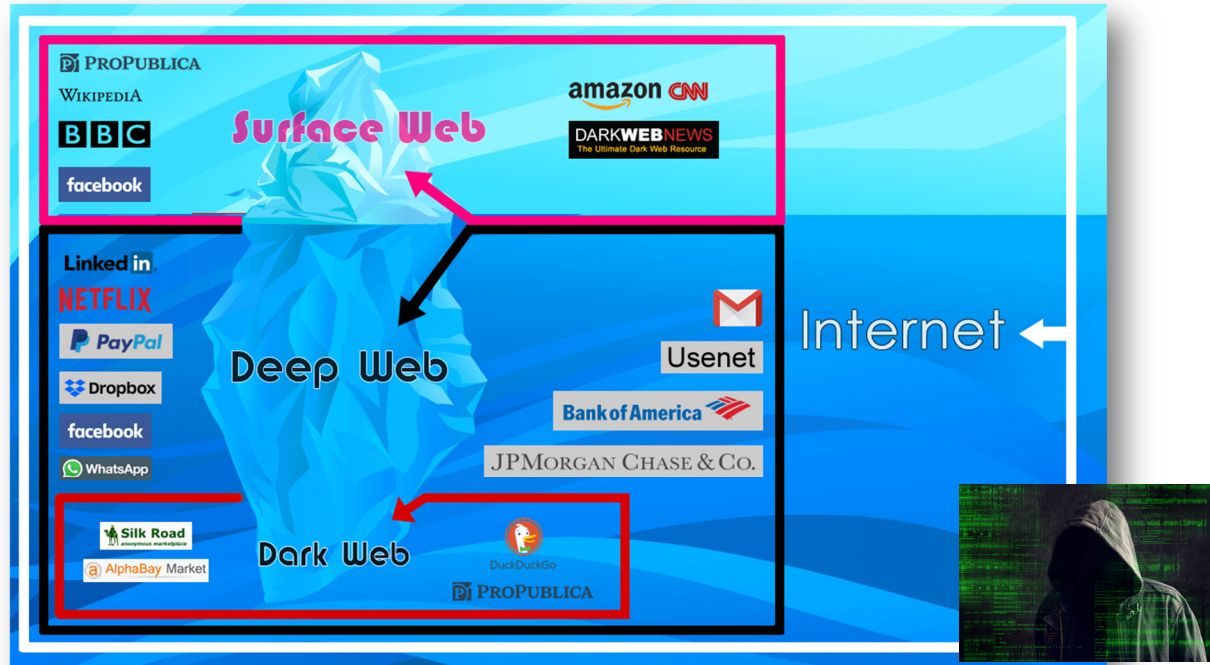
Any web page that is not indexed by a search engine (e.g., dynamic pages, password-protected content, ...)

Dark Web

- Part of the Deep Web that is not just hidden from search engines, but is specifically designed to be inaccessible from most web browsers (it requires special browsing software, which makes it much harder to track the path data takes online; e.g., the TOR — The Onion Router — browser, <https://www.torproject.org/>)
- Mostly used for illegal purposes, but also to bypass censorship



The “other” Web ... *Deep and Dark Web*...



[illegible]

“Automatic” creation of websites

Content Management Systems

WCMS: Web Content Management System

Software (integrated environment) that allows the creation and management of websites

Generally, when we talk about CMS, we mean WCMS

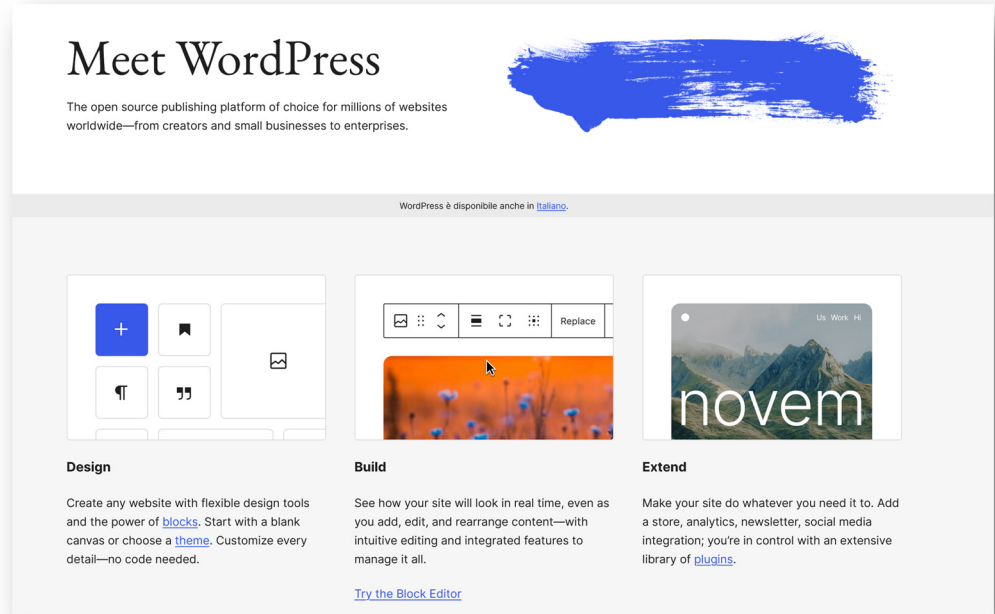
- CMSs support server-side database interaction
- Dynamic websites can be built (almost) without writing HTML, CSS, JavaScript, PHP, etc.

The most popular open-source CMSs

WordPress

Initially born as a
blogging platform

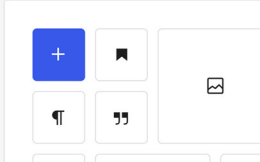
wordpress.org



Meet WordPress

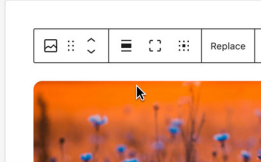
The open source publishing platform of choice for millions of websites worldwide—from creators and small businesses to enterprises.

WordPress è disponibile anche in [Italiano](#).



Design

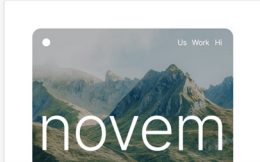
Create any website with flexible design tools and the power of [blocks](#). Start with a blank canvas or choose a [theme](#). Customize every detail—no code needed.



Build

See how your site will look in real time, even as you add, edit, and rearrange content—with intuitive editing and integrated features to manage it all.

[Try the Block Editor](#)



Extend

Make your site do whatever you need it to. Add a store, analytics, newsletter, social media integration; you're in control with an extensive library of [plugins](#).

The most popular open-source CMSs

Joomla

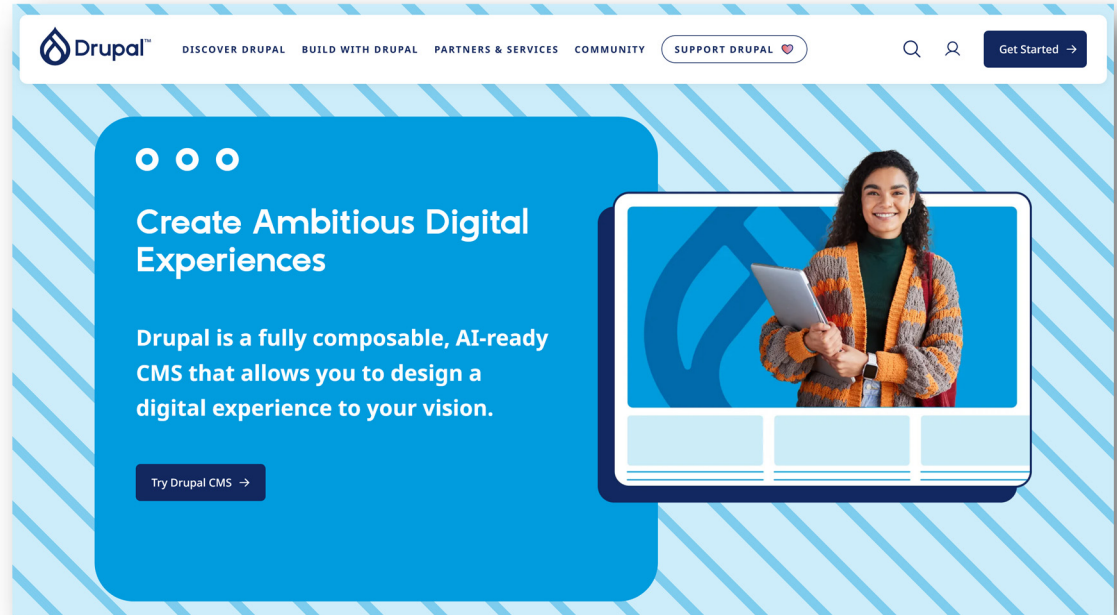
joomla.org



The most popular open-source CMSs

Drupal

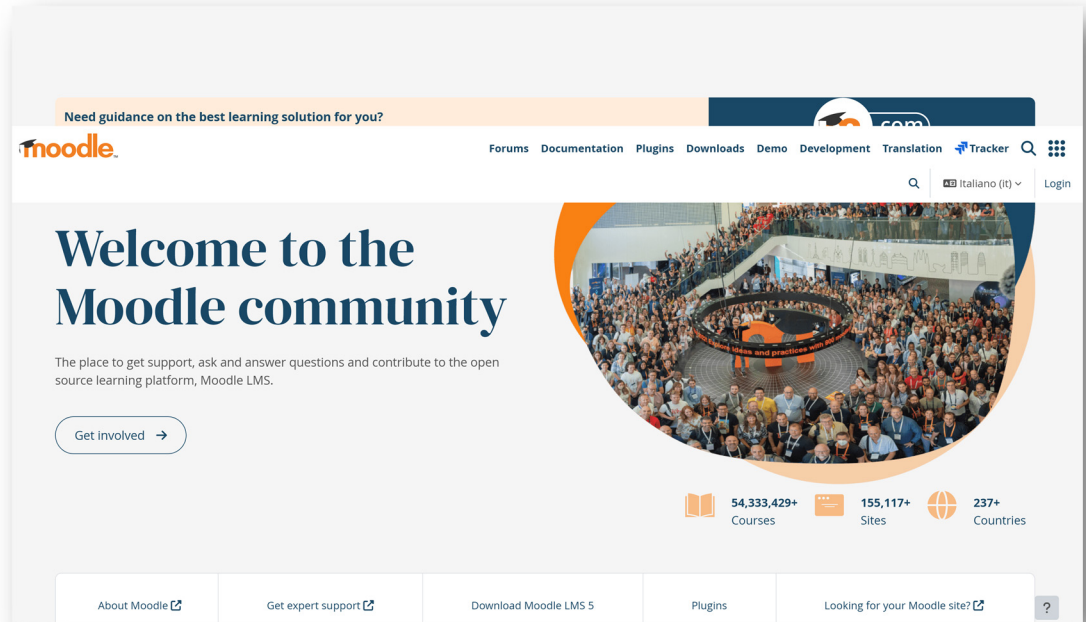
drupal.org



The most popular open-source CMSs

Moodle

moodle.org



Smartphones: Web Apps or Apps?

Web app

RIA (rich internet application) run in the smartphone's browser

App

Program natively run by the smartphones processor

... usually, applications are more powerful and customizable

